LifeWise — Complete Technology Stack

Intelligent Life Insurance Needs Finder & Digital Advisory Platform

TalentAP Mega Hackathon 2026 • Problem B2 (BFSI) • Authoritative Architecture Document

Architecture Summary: LifeWise implements a high-performance, deterministic full-stack architecture. The platform decouples actuarial calculation engines from presentation, strictly grounding all quotes and underwriting in supplied synthetic datasets with zero external web lookups or fabricated premiums. Below is the complete layer-by-layer technical specification.

1. Frontend Architecture & Client Technologies

Layer / Tool	Technology	Version	Role in LifeWise Platform
Core Framework	React	18.3.1	Modular, component-based single page application
Language	TypeScript	5.5.3	Strict static typing for financial models & schemas
Build Engine	Vite (Rolldown)	5.4.2	Sub-second HMR & production bundle optimization
UI & Styling	Tailwind CSS	3.4.1	Enterprise fintech design system, responsive grids
Data Viz	Recharts	2.12.7	Interactive waterfall, donut, and timeline charts
Icons	Lucide React	0.344.0	Consistent financial and status iconography
Offline Fallback	OfflineEngine	Native	Zero-dependency client calculation fallback

2. Backend Architecture & API Layer

Layer / Tool	Technology	Version	Role in LifeWise Platform
API Framework	FastAPI	0.110.0	High-throughput async REST API endpoints
Web Server	Uvicorn	0.28.0	ASGI production web server on port 8000
Data Validation	Pydantic v2	2.6.4	Strict input sanitization & financial response contracts
ORM / DB Layer	SQLAlchemy	2.0.28	Relational model mapping & persistence
Embedded DB	SQLite	3.x	Zero-dependency embedded relational storage
Settings Manager	pydantic-settings	2.2.1	12-factor environment variable management
Runtime	Python	3.10+	Standard library Decimal for round-half-up math

3. Deterministic Actuarial Engines & Dataset Grounding

Engine Module	Implementation Class	Data Source	Mathematical / Regulatory Methodology
Capital Needs Analysis	FinancialCalculationEngine	customer_profiles.json	Max(0, Income PV + Debt + Goals PV + Reserve - Assets - Cover)
Human Life Value (HLV)	FinancialCalculationEngine	reference_assumptions	Fisher real discount rate: (1 + i)/(1 + r) - 1 with personal share deduction
Tabular Premium Engine	PremiumCalculationEngine	premium_rates.csv	39,588 row lookup: Rate x (SA/1000) x Band Factor x Modal Factor (0.0875)
Rider Pricing & Caps	PremiumCalculationEngine	riders.csv	Statutory cap: Health riders <= 100%, other riders <= 30% of base

ULIP Simulator	PremiumCalculationEngine	ulip_funds.csv, charges	IRDAI 4% & 8% benchmarks, monthly attained-age mortality deductions
7-Dimension Risk Audit	PolicyRiskIntelligenceEngine	products.json, rates	Financial, coverage, eligibility, structure, rider, ULIP, completeness
Reverse Budget Solver	PolicyRiskIntelligenceEngine	premium_rates.csv	Reverse solves max affordable sum assured matching budget X/month

4. Automated Testing & Verification Suite

• **Pytest Automated Suite:** 47 / 47 tests passing (100% success rate in 1.1s).
• **Coverage:** Worked example exact match (■1,045/mo), decimal halves round-up, homemaker economic valuation, age and maturity limits, underwriting caps, statutory rider limits, and slot collision prevention.
• **Frontend Compilation:** Strict TypeScript build (0 errors) using Vite.